

PHYSICS

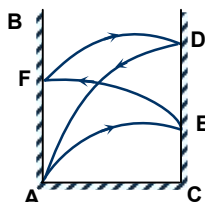
TARGET : JEE- Advanced

CAPS-7

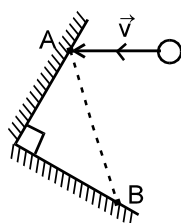
CENTRE OF MASS

SCQ (Single Correct Type) :

1. A particle is projected from the foot of one wall towards other parallel wall as shown in the figure. After various collisions, finally after striking at point D it comes back directly to the point of projection. If 'e' is coefficient of restitution for each impact, then $e^3 + e^2 + e$ is equal to

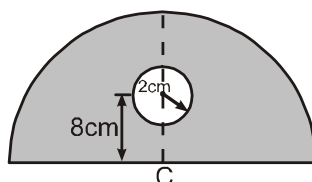


- (A) 1 (B) 2 (C) 1.5 (D) 2.5
2. AB is an L shaped obstacle fixed on a horizontal smooth table. A ball strikes it at A, gets deflected and restrikes it at B. If the velocity vector before collision is $\vec{v}$ and coefficient of restitution of each collision is 'e', then the velocity of ball after its second collision at B is



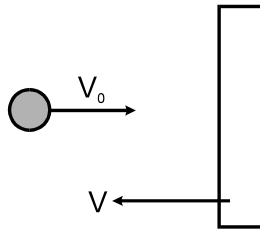
- (A) $e^2\vec{v}$ (B) $-e^2\vec{v}$ (C) $-e\vec{v}$ (D) data insufficient
3. A continuous stream of particles of mass m and velocity v , is emitted from a source at a rate of n per second. The particles travel along a straight line, collide with a body of mass M and are buried in this body. If the mass M was originally at rest, its velocity when it has received N particles will be:

- (A) $\frac{mvn}{Nm+n}$ (B) $\frac{mvN}{Nm+M}$ (C) $\frac{mv}{Nm+M}$ (D) $\frac{Nm+M}{mv}$
4. In the figure shown a hole of radius 2 cm is made in a semicircular disc of radius 6π cm at a distance 8 cm from the centre C of the disc. The distance of the centre of mass of this system from point C is:



- (A) 4 cm (B) 8 cm (C) 6 cm (D) 12 cm

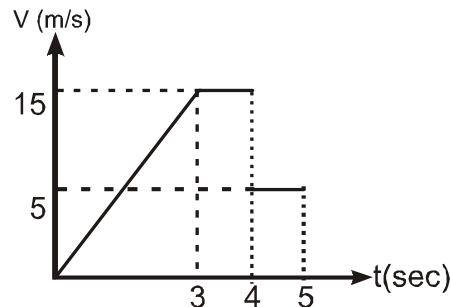
5. A plate, of uniform thickness and uniform density, has shape in the x-y plane defined by the lines $x = 0$, $y = 2$, and curve $y = \frac{1}{2}x^2$. The plate lies in first quadrant of x-y plane. Then the x-coordinate of centre of mass of this plate is:
- (A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $\frac{4}{3}$ (D) $\frac{3}{4}$
6. A block of mass $m = 3\text{ kg}$ resting on a horizontal frictionless floor is horizontally struck by a 9N force that acts for 0.02 sec. After 3 sec it receives a second blow of force 9N but in opposite direction which acts for 0.01 sec. The speed of the body after 30 sec. is
- (A) 0 (B) 3 cm/sec (C) 90 cm/sec (D) 30 cm/sec
7. A particle of mass m moves with velocity $v_0 = 20\text{ m/sec}$ towards a wall that is moving with velocity $v = 5\text{ m/sec}$. If the particle collides elastically with the wall, the speed of the particle just after the collision is :



- (A) 30 m/s (B) 20 m/s (C) 25 m/s (D) 22 m/s

MCQ (One or more than one correct) :

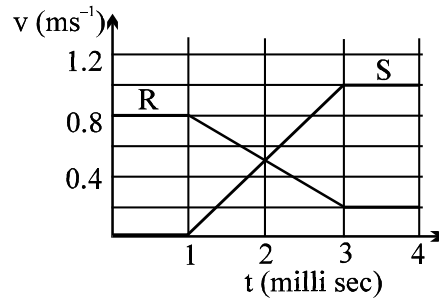
8. The figure shows the velocity as a function of the time for an object with mass 10 kg being pushed along a frictionless horizontal surface by external horizontal force. At $t = 3\text{ s}$, the force stops pushing and the object moves freely. It then collides head on and sticks to another object of mass 25 kg.



- (A) External force acting on the system is 50 N from $t = 0$ to $t = 3\text{ sec}$.
 (B) Speed of the 2nd particle just before the collision is 1 m/s.
 (C) before collision both bodies are moving in the same direction.
 (D) before collision, bodies are moving in opposite direction.
9. The diagram to the right shows the velocity-time graph for two masses R and S that collided elastically. Which of the following statements is true?
- (I) R and S moved in the same direction after the collision.

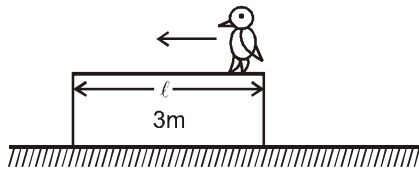
(II) Kinetic energy of the system (R & S) is minimum at $t = 2$ milli sec.

(III) The mass of R was greater than mass of S.

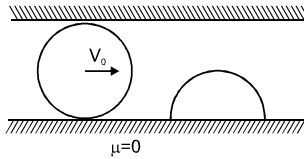


- (A) I only (B) II only (C) I and II only (D) I, II and III

10. A penguin of mass m stands at the right edge of a sled of mass $3m$ and length ℓ . The sled lies on frictionless ice. The penguin starts moving towards left, reaches the left end and jumps with a velocity u and at an angle θ relative to ground. (Neglect the height of the sled)



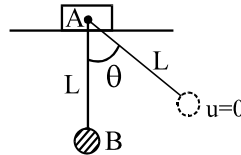
- (A) Till the penguin reaches the left end, the sled is displaced by $\frac{\ell}{4}$
- (B) Till the penguin reaches the left end, the sled is displaced by $\frac{\ell}{3}$
- (C) After jumping, it will fall on the ground at a distance $\frac{4}{3} \frac{u^2 \sin 2\theta}{g}$ from the left end of the sled.
- (D) After jumping, it will fall on the ground at a distance $\frac{3}{4} \frac{u^2 \sin 2\theta}{g}$ from the left end of the sled.
11. A particle moving with kinetic energy = 3 J makes an elastic head-on collision with a stationary particle which has twice its mass. During the impact :
- (A) the minimum kinetic energy of the system is 1 J
- (B) the maximum elastic potential energy of the system is 2 J
- (C) momentum and total energy are conserved at every instant
- (D) the ratio of kinetic energy to potential energy of the system first decreases and then increases.
12. A smooth hemisphere of mass m and radius R is at rest. A smooth solid sphere of mass $2m$ and radius R moving with velocity V_0 between two horizontal smooth surfaces separated by a distance slightly greater than $2R$ as shown in figure. Solid sphere collides with the hemisphere. If coefficient of restitution is $\frac{1}{2}$, then :



- (A) The speed of hemisphere after collision is V_0
- (B) The speed of solid sphere after collision is $\frac{V_0}{2}$
- (C) The loss in kinetic energy of the system is $\frac{1}{4} mV_0^2$
- (D) The final kinetic energy of hemisphere is $\frac{1}{4}$ th the initial kinetic energy of sphere

Comprehension Type Question:

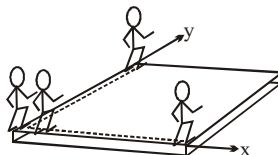
A small ball B of mass m is suspended with light inelastic string of length L from a block A of same mass m which can move on smooth horizontal surface as shown in the figure. The ball is displaced by angle θ from lower most position & then released.



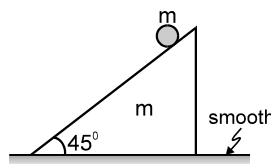
13. The displacement of block when ball reaches the lower most position is
- (A) $\frac{L \sin \theta}{2}$ (B) $L \sin \theta$ (C) L (D) none of these
14. Maximum velocity of block during subsequent motion of the system after release of ball is
- (A) $[gl(1 - \cos \theta)]^{1/2}$ (B) $[2gl(1 - \cos \theta)]^{1/2}$
- (C) $[gl \cos \theta]^{1/2}$ (D) informations are insufficient to decide
15. The displacement of centre of mass of A + B system till the string becomes vertical is
- (A) zero (B) $\frac{L}{2}(1 - \cos \theta)$ (C) $\frac{L}{2}(1 - \sin \theta)$ (D) none of these

Numerical based Questions :

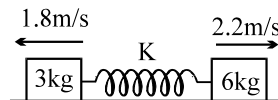
16. A square plank of mass $m_1 = 100$ kg and edge length $L = 20\sqrt{2}$ m is placed on a smooth surface. Two person each of mass $m_2 = m_3 = 50$ kg are at corner of a plank as shown in figure. Two person begin to walk on the plank along two different paths as shown in figure and reach nearest corners. What is magnitude of displacement of plank (in m) in the process.



17. Two balls moving on two parallel lines $x = a$ and $x = 2a$ undergo on elastic collision with a parabolic surface $x^2 = 4ay$. Find the Y-coordinates of the particles from which they will pass after collision.
18. A 45.0-kg girl is standing on a plank that has a mass of 150 kg. The plank, originally at rest, is free to slide on a frozen lake, which is a flat, frictionless supporting surface. The girl begins to walk along the plank at a constant speed of 1.50 m/s relative to the plank. (a) What is her speed relative to the ice surface? (b) What is the speed of the plank relative to the ice surface?
19. A ball of mass 'm' is released from the top of a smooth movable wedge of mass m as shown. When the ball collides with the horizontal floor, velocity of the wedge is $v = 10$ m/s. Find the maximum height (in meter) attained by the ball above the floor after an elastic collision with the floor. (Take $g = 10 \text{ m/s}^2$)

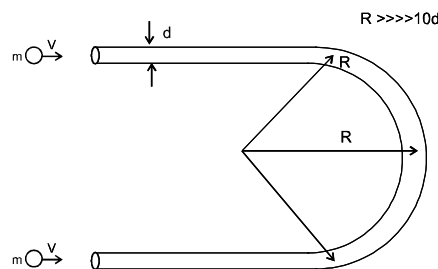


20. Two blocks A & B are connected by a spring of stiffness 512 N/m and placed on a smooth horizontal surface. Initially the spring has its equilibrium length. The indicated velocities are imparted to A & B. Find the maximum extension in the spring ?



Matrix Match Type :

21. A U shaped tube of mass 2m is placed on a horizontal surface. Two identical spheres each of diameter d (just less than the inner diameter of tube) and mass m enter into the tube with velocity v as shown in the figure taking all collisions to be elastic and all surfaces smooth.



Column-I

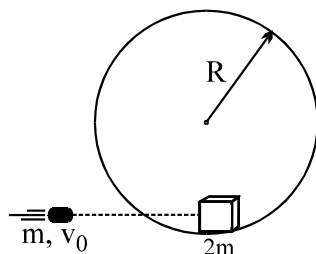
- (A) The speed of the tube.
when spheres are just about to collide inside the tube.
- (B) The speed of spheres when spheres are just about to collide.
- (C) The speed of the spheres when they come out of the tube.
- (D) The speed of the tube when spheres come out

Column-II

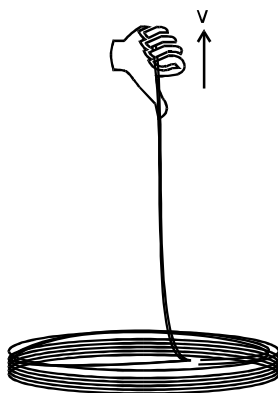
- (p) V
- (q) 0
- (r) $\frac{\sqrt{3}}{2} V$
- (s) $v/2$

Subjective Type Questions :

22. A small block of mass $2m$ initially rests at the bottom of a fixed circular, vertical track, which has a radius of R . The contact surface between the mass and the loop is frictionless. A bullet of mass m strikes the block horizontally with initial speed v_0 and remain embedded in the block as the block and the bullet circle the loop. Determine each of the following in terms of m , v_0 , R and g .



- (a) The speed of the masses immediately after the impact.
 (b) The minimum initial speed of the bullet if the block and the bullet are to successfully execute a complete ride on the loop
23. A uniform rope of linear mass density λ and length ℓ is coiled on a smooth horizontal surface. One end is pulled up with constant velocity v . Then find average power applied by the external agent in pulling the entire rope just off the ground ?



24. A ball moving translationally collides with another stationary ball of the same mass. At the moment of impact the angle between the straight line passing through the centres of the balls and the direction of the initial motion of the striking ball is equal to $\alpha = 45^\circ$. Assuming the balls to be smooth, find the fraction η of the kinetic energy of the striking ball that turned into potential energy at the moment of the maximum deformation.
25. A sphere of mass m is moving with a velocity $4\hat{i} + 3\hat{j} - 5\hat{k}$ when it hits a smooth wall and rebounds with velocity $2\hat{i} + 2\hat{j} + 3\hat{k}$. Find the impulse it receives. Find also the coefficient of restitution between the sphere and the wall.

ANSWER KEY (CAPS-7)

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|--|---|---|---|----------|
| 1. (A) | 2. (C) | 3. (B) | 4. (B) | 5. (D) |
| 6. (B) | 7. (A) | 8. (ABC) | 9. (D) | 10. (AC) |
| 11. (ABCD) | 12. (ABC) | 13. (A) | 14. (A) | 15. (B) |
| 16. 10 | 17. a | 18. (a) 15/13 m/s (b) -9/26 m/s] | | |
| 19. $\frac{2v^2}{g} = 20$ | 20. 25 cm | 21. (A) - s ; (B) - r ; (C) - q ; (D) - p | | |
| 22. (a) $v_0/3$, (b) $3\sqrt{5gR}$ | 23. $\frac{\lambda \ell}{2} \frac{vg}{2} + \lambda v^3$ | | 24. $\eta = \frac{1}{2} \cos^2 \alpha = 0.25$ | |
| 25. $m(-2\hat{i} - \hat{j} + 8\hat{k})$, $e = \frac{6}{17}$ | | | | |